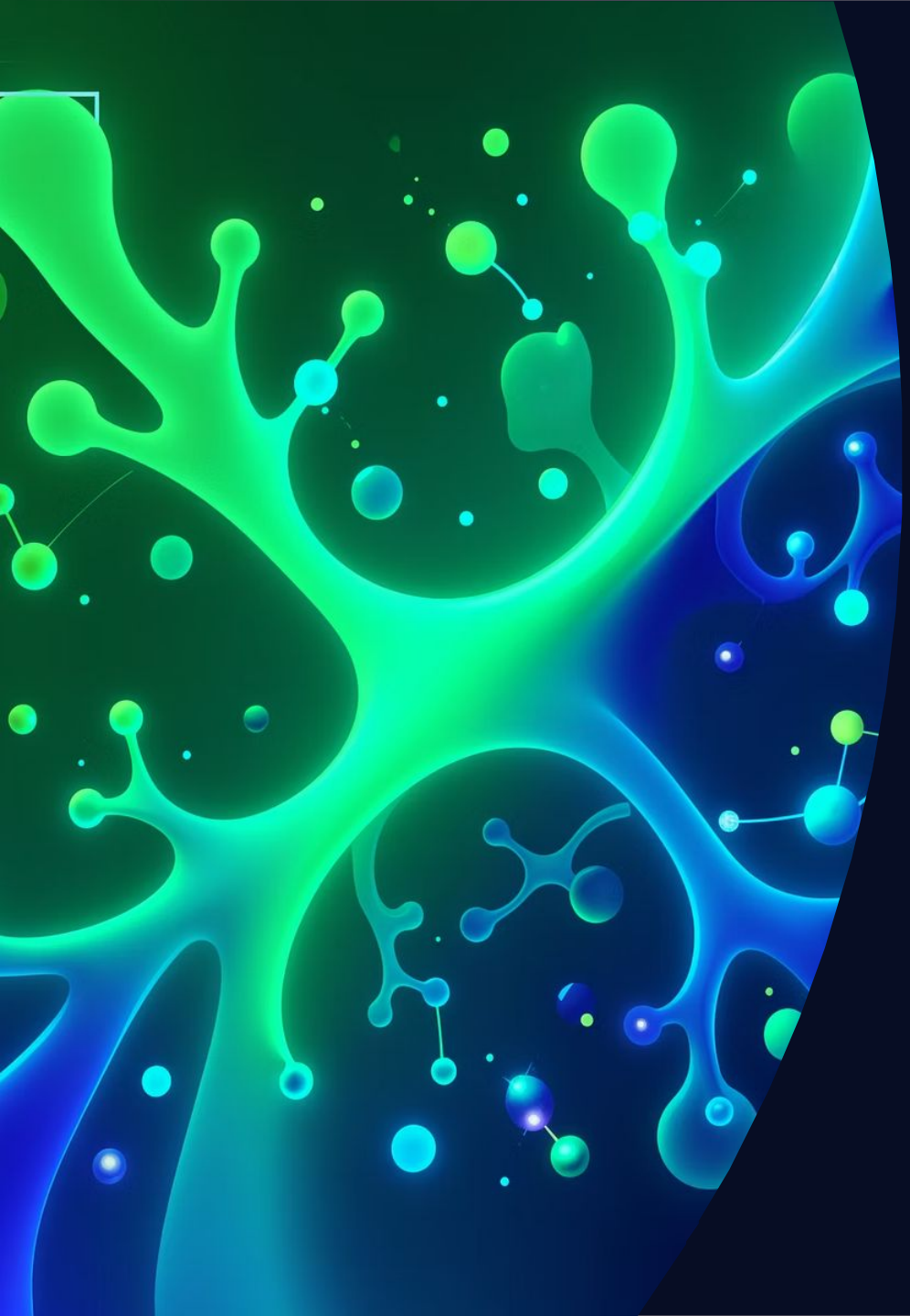


STOCHASTIC SIMULATION

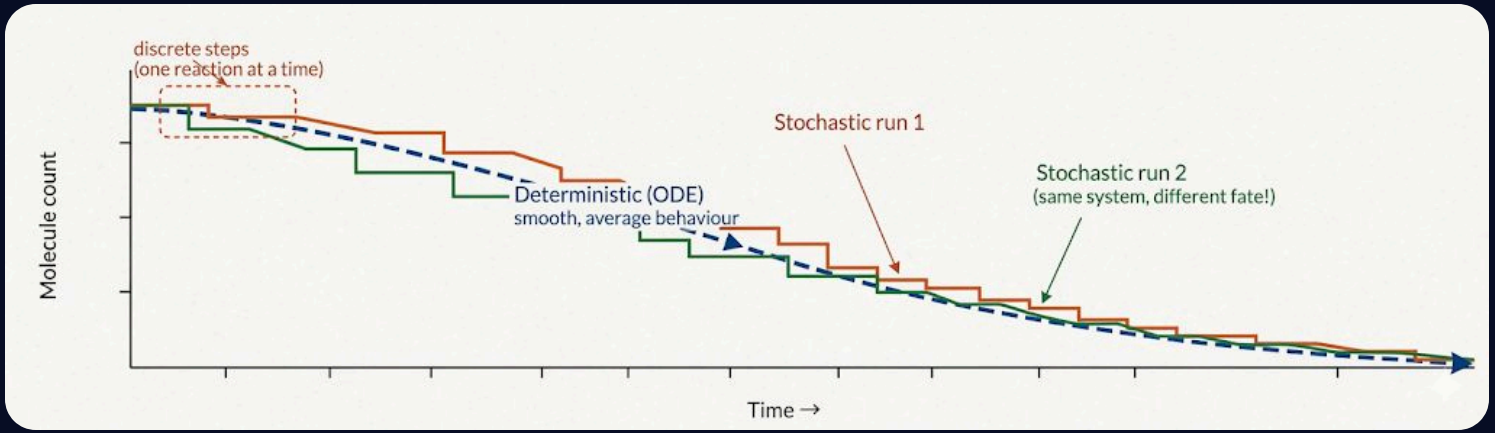
Gillespie Algorithm

by Saswata Chatterjee

The Gillespie Algorithm captures the unpredictable rhythm of chemical reactions by sampling each event in real time—revealing how randomness shapes dynamics one precise stochastic step at a time.

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Why Stochastic & Exact Simulation?



This image contrasts a smooth, deterministic ordinary differential equation (ODE) trajectory with two discrete, fluctuating stochastic simulations of the exact same chemical system. It visually demonstrates how individual stochastic runs capture raw microscopic randomness and can lead to different trajectories over time, whereas the deterministic model only tracks the macroscopic average behavior

Deterministic vs. Stochastic

Classical ODEs

Assume smooth, predictable concentration changes.

Stochastic Reality

Captures random thermal collisions and event timing.

Discrete Tracking

Models integer molecule counts: 1, 2, 3...

The Meaning of "Exact"

No Time Approximations

Avoids fixed-step errors from Δt-based methods.

True Event Intervals

Computes the exact random time between events.

No Averages

Simulates raw randomness without smoothing or simplification.

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Mathematical Foundations of the SSA

Three core definitions determine how the Stochastic Simulation Algorithm models time, events, and molecular counts.

1. System State

The state vector X(t) stores the integer molecule counts for every species at time t. It gets updated everytime a reaction occurs with respect to the afforementioned specie of molecules.

2. Propensity Functions

Each reaction has a propensity a_i, the probability rate that it fires in the next instant.

$$a_i = c_i X_i$$

c_i represents the stochastic rate constant for species iX_i is the number of molecules of species i.

3. Total Propensity

The system propensity a_0 is the sum of all reaction propensities.

$$a_0 = \sum_{i=1}^{M} a_i$$

M is the total number of directed pathways.

4. Probability distribution

- **Binomial origin**: Start with N molecules, each with a small reaction probability p in a time step.
- **Poisson limit**: As $N \to \infty$ and $p \to 0$, while $\lambda = Np$ stays constant, the event count approaches a Poisson distribution. We can hence obtain the **PDF** for the distribution:

$$P(\tau) = \alpha_0 \cdot e^{-\alpha_0 \tau}, \quad \tau \geq 0$$

Here τ is defined as the waiting time till the next reaction takes place,

- **The CDF**:

$$F(\tau) = 1 - e^{-\alpha_0 \tau}$$

For the exponential distribution used in the Gillespie algorithm, the CDF is obtained by integrating the PDF from 0 to τ.

5. Waiting time (τ)

- **Definition**: A continuous random variable τ representing the exact time elapsed until the next discrete reaction event.
- **Distribution**: Governed by the **Exponential Distribution** with rate parameter α_0 (total system propensity).
- **Key Property**: **Memoryless** — past idle time does not influence future collision probabilities.
- **Sampling Inversion**: Computed instantly from a uniform random float $r_1 \in (0, 1)$ via:

$$\tau = \frac{1}{\alpha_0} \ln\left(\frac{1}{r_1}\right)$$

6. Reaction Selection:

- **The Method**: Maps a uniform random number $r_2 \in (0, 1)$ onto a discrete probability line where each segment's length is proportional to its reaction propensity $r_2 \in (0, 1)$.
- **The Scale**: Multiplying $r_2 \cdot \alpha_0$ scales the random number from a (0,1) range to a $(0, \alpha_0)$ range.
- **The Selection Law**: Find the unique index j such that:

$$\sum_{i=1}^{j-1} \alpha_i(\mathbf{X}) < r_2 \alpha_0(\mathbf{X}) \leq \sum_{i=1}^{j} \alpha_i(\mathbf{X})$$

- **State Update**: Once j is identified, the system state is instantly updated using its stoichiometric vector: $\mathbf{X} \to \mathbf{X} + \nu_j$

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Algorithmic Flow of the Gillespie Algorithm

A dual-clock simulation engine advances stochastic reactions while independently recording observations at fixed intervals.

1. Parameter Initialization

Set the simulation parameters and prepare the trajectory buffer for ensemble tracking.

- Rate constants: $k_1, k_2, \ldots, k_M$
- Maximum simulation window: m_t
- Observation interval: Δt
- Tracking matrix: $\text{conc} \in \mathbb{R}^{I \times J}$

2. Total Propensity Computation

Evaluate the reaction hazards for the current molecular state and aggregate them into the total system propensity.

$\mathbf{X}$ is the current molecular counts vector, with individual propensities $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_M$.

$$a_0 = \sum_{m=1}^{M} \alpha_m = \alpha_1 + \alpha_2 + \alpha_3 + \cdots + \alpha_M$$

3. Stochastic Event Advance

Sample the next reaction time from the exponential waiting-time distribution and advance the stochastic clock.

Draw a uniform random value $r_1 \in (0, 1)$ and compute the waiting time τ.

$$\tau = \frac{1}{a_0} \ln\left(\frac{1}{r_1}\right)$$

Update the reaction timeline:

$$t_1 \leftarrow t_1 + \tau$$

4. On-the-Fly Grid Synchronization

Align the stochastic reaction clock t_1 with the uniform observation clock t_2 and record any completed observation points before the reaction fires.

While $t_2 + \Delta t \le t_1$, advance the reporting grid and store the current pre-reaction state.

- $t_2 \leftarrow t_2 + \Delta t$
- $j \leftarrow j + 1$
- $\text{conc}[i, j] = \mathbf{X}$

5. State Vector Selection & Mutation

Use a second uniform random number to select the next reaction, then update the molecular state using its stoichiometric change vector.

Draw $r_2 \in (0, 1)$ and identify the unique reaction index m satisfying the cumulative propensity partition.

$$\sum_{m'=1}^{m-1} \alpha_{m'}(\mathbf{X}) < r_2 a_0(\mathbf{X}) \le \sum_{m'=1}^{m} \alpha_{m'}(\mathbf{X})$$

$$\mathbf{X} \leftarrow \mathbf{X} + \nu_m$$

6. Boundary Padding & Ensemble Reductions

Terminate when the simulation reaches the maximum time horizon, then finalize any missing observation entries and summarize the ensemble.

- If $t_1 \ge m_t$, stop the execution loop.
- Fill any remaining empty tracking columns with the final state.
- Repeat for I independent runs.
- Compute ensemble statistics with `np.mean(conc, axis=0)` and `np.var(conc, axis=0)`.



The result is a synchronized stochastic workflow that preserves exact reaction timing while producing uniformly sampled trajectory summaries.

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Case Study: The Single Degradation Reaction

A minimal stochastic system that illustrates how a single reaction channel drives molecular decay over time.

1. The Model System

One species, A, degrades irreversibly at a constant rate.

$$A \xrightarrow{k_1} \emptyset$$

The discrete state is the molecule count of the species:

$$\mathbf{X}(t) = [A(t)]$$

2. Propensity & System Rates

The reaction propensity scales linearly with the available molecules.

$$\alpha_1(A) = k_1 \cdot A$$

Because there is only one active pathway, the total system propensity is the same as the single reaction propensity:

$$a_0 = \sum_{m=1}^{1} \alpha_m = \alpha_1(A) = k_1 \cdot A$$

3. Understanding the Kinetic Trajectory

Each firing event removes one molecule from the system.

$$\nu_1 = [-1] \implies A \leftarrow A - 1$$

The waiting time between reaction events is sampled from an exponential distribution:

$$\tau = \frac{1}{a_0} \ln\left(\frac{1}{r_1}\right)$$

As A decreases, a_0 also decreases, so successive reaction intervals become progressively longer as the system approaches exhaustion.

This case study provides a clean reference for understanding how stochastic reaction timing, discrete state updates, and propensity-driven dynamics interact in a single-reaction system.

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STOCHASTIC SIMULATION GILLESPIE

Slide Summary

A stochastic view of molecular degradation highlights how random reaction timing shapes both individual trajectories and population-level behavior.

1. Individual Trajectories

Colored lines show discrete, jagged step functions from single stochastic runs of molecular degradation $A \xrightarrow{k} \emptyset$.

This captures distinct integer events, with each reaction producing $\Delta A = -1$ after a probabilistic waiting time.

2. Ensemble Mean

The solid black line tracks the average molecule count across all runs, represented by $\langle A(t) \rangle$.

Despite microscopic randomness, the mean follows the deterministic rate law $[A](t) = [A]_0 e^{-kt}$.

This can be derived both from the Chemical Master equation (CME).

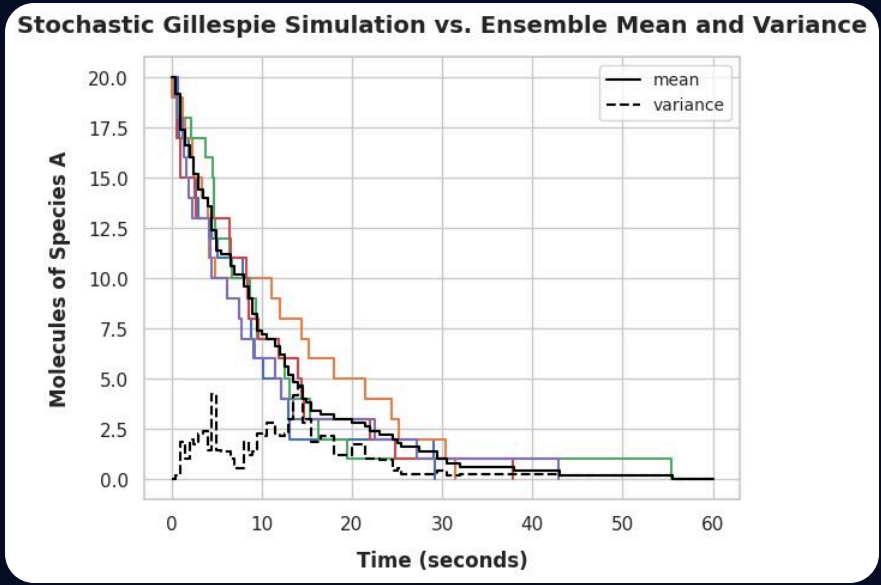
3. Ensemble Variance

The dashed black line shows the spread across trajectories over time.

At $t = 0$: variance is 0 because all trajectories begin at exactly $N = 20$.

Mid-phase ($t \approx 2$–$12\,\text{s}$): variance peaks as random timing maximizes divergence.

Late phase ($t > 30\,\text{s}$): variance returns to 0 as trajectories reach the absorbing state $A = 0$.



Core Takeaway

The ensemble mean matches perfectly with the values derived from the steady state equations solved deterministically.

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Case Study: The Production–Degradation Network

A linear birth-death system illustrates how constant production and proportional degradation combine to create a dynamic steady state with a Poisson-distributed molecule count.

1. The Model Framework

The Reaction Network: Extending simple decay into an open system by coupling constant production with linear population degradation:

$$A \xrightarrow{k_1} \emptyset \quad (\text{Degradation})$$

$$\emptyset \xrightarrow{k_2} A \quad (\text{Production})$$

Discrete State System: The active variable tracks the exact integer population of a single species over time, with initial condition $X(t) = [A(t)]$ and $A(0) = 0$.

2. Dual-Propensity Formulation

The system pace is governed by two competing reaction channels that combine into a single dynamic clock variable a_0.

Degradation Propensity: $\alpha_1(A) = k_1 \cdot A$

Production Propensity: $\alpha_2(A) = k_2$

Total System Propensity :

$a_0 = \sum_{m=1}^{2} \alpha_m = k_1 A + k_2$

3. Kinetic Profile Analysis

Relaxation Phase ($t = 0 \to 30$s): All trajectories start identically at $A = 0$. Early on, $a_0 \approx k_2$, producing rapid back-to-back jumps $A \leftarrow A + 1$ before degradation increasingly shapes the variance.

Stationary Phase ($t = 30 \to 60$s): The ensemble mean and variance flatten, indicating that the system has reached a dynamic steady-state equilibrium.

4. Stationary Probability Distribution

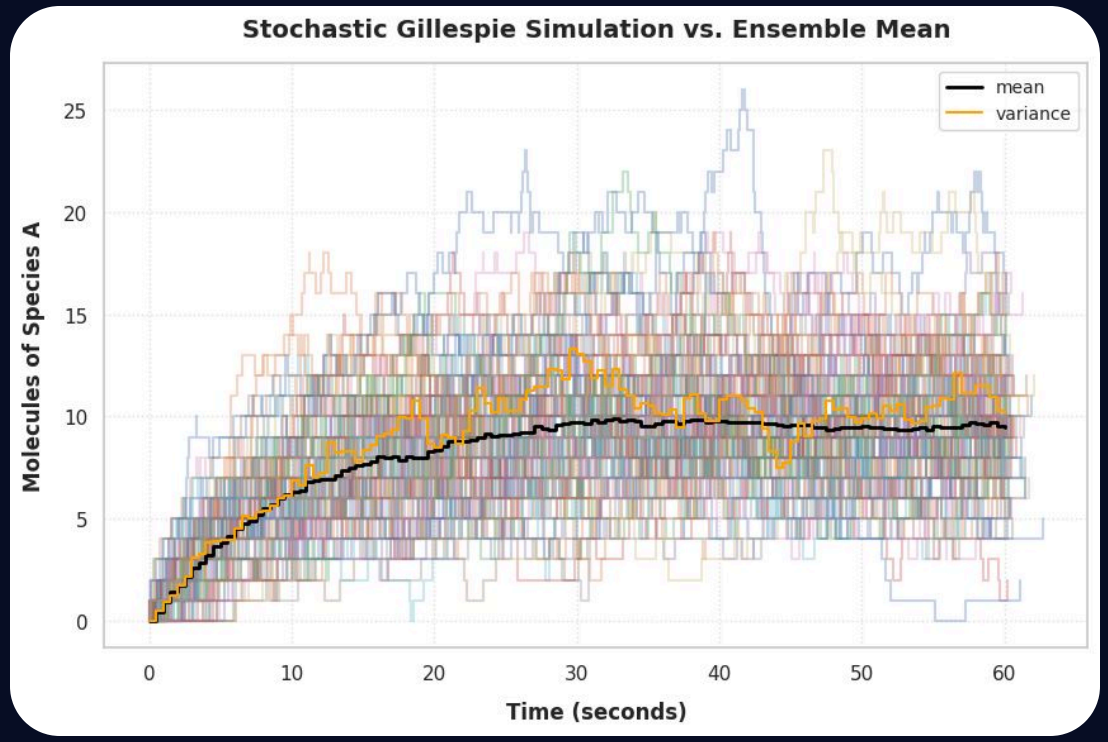
The Distribution Curve: Extracting terminal values at $t = 60$s across all 100 trajectories yields a discrete distribution centered near 10 molecules.

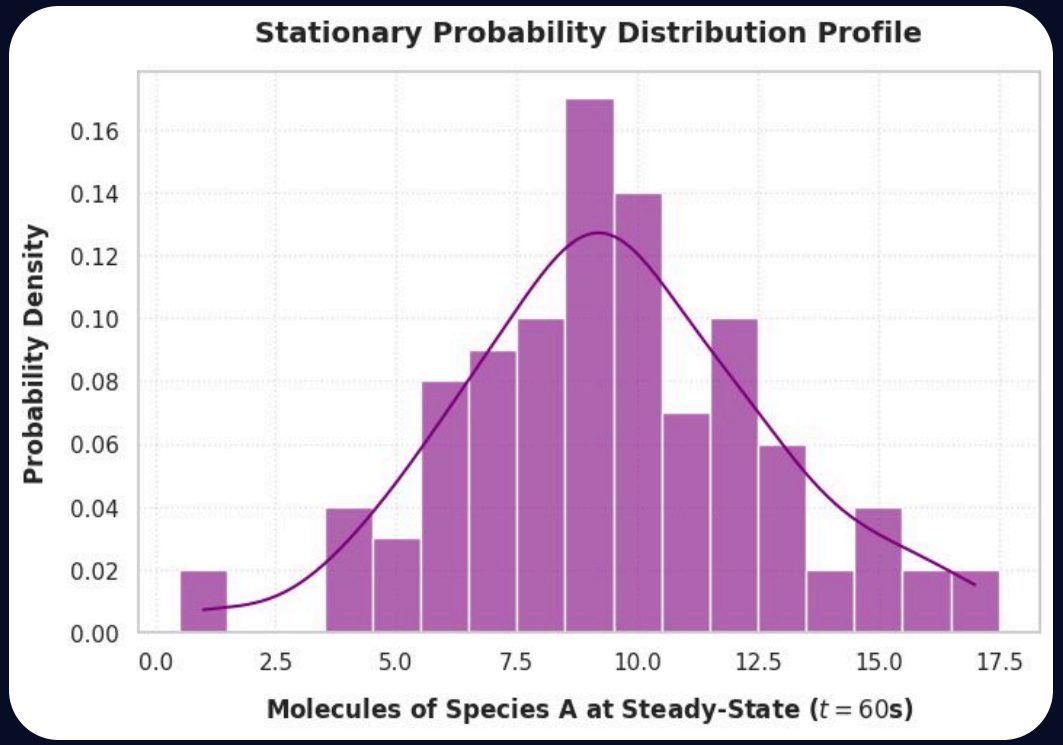
The Poisson Limit: Because the system is linear, the Chemical Master Equation converges to a Poisson distribution:

$$P_{ss}(A) = \frac{\left(\frac{k_2}{k_1}\right)^A e^{-\frac{k_2}{k_1}}}{A!}$$

Statistical Validation: The histogram confirms the Poisson constraint, with $\langle A \rangle_{ss} \approx \mathrm{Var}(A)_{ss} \approx 10$.

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The fundamental conclusion is that while individual molecular trajectories are highly stochastic and noisy, the ensemble statistics reveal deep, predictable macroscopic limits where the physical architecture of the chemical network dictates its stationary behavior. For the linear production-degradation network, the convergence of both the ensemble mean and variance to the exact same value ≈ 10 provides solid statistical validation that the system reaches a steady-state equilibrium governed strictly by a Poisson distribution.

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Case Study: The Dimerization–Production Network

How nonlinear collision-driven decay creates a tightly regulated, non-Poissonian steady state.

The Model Framework

The network combines constant synthesis with second-order dimerization loss, making the dynamics fundamentally nonlinear.

$$A + A \xrightarrow{k_1} \emptyset \quad \text{(Dimerization Decay)}$$

$$\emptyset \xrightarrow{k_2} A \quad \text{(Constant Influx)}$$

The state space tracks the discrete copy number of A, and each dimerization event removes exactly two molecules:

$$\nu_1 = [-2] \implies A \leftarrow A - 2$$

Combinatorial Propensity Formulation

Because degradation requires a collision between two identical molecules, the reaction hazard grows quadratically with copy number.

$$\alpha_1(A) = k_1 \frac{A(A-1)}{2}$$

$$\alpha_2(A) = k_2$$

$$a_0 = \sum_{m=1}^{2} \alpha_m = k_1 \frac{A(A-1)}{2} + k_2$$

This combinatorial structure is the source of the strong self-limiting feedback in the system.

Stationary Probability Distribution & The CME

The Chemical Master Equation governs the time evolution of the probability of state n:

$$\frac{dP_n}{dt} = k_1 \frac{(n+2)(n+1)}{2} P_{n+2} - k_1 \frac{n(n-1)}{2} P_n + k_2 P_{n-1} - k_2 P_n$$

At steady state, inflows and outflows balance exactly, producing a stationary probability function $P(n)$:

$$k_1 \frac{(n+2)(n+1)}{2} P(n+2) + k_2 P(n-1) = \left[k_1 \frac{n(n-1)}{2} + k_2 \right] P(n)$$

The simulated histogram peaks near 14 molecules, validating the numerical steady-state distribution generated by the Gillespie algorithm.

Non-Poissonian Behavior: Mean vs. Variance

Moment analysis shows that the steady-state fluctuations are suppressed below a Poisson baseline.

$$\mathrm{Var}(n)_{ss} = \frac{3k_2}{2k_1(2\langle n \rangle - 1)}$$

$$\mathrm{Var}(n)_{ss} \approx 0.75 \cdot \langle n \rangle$$

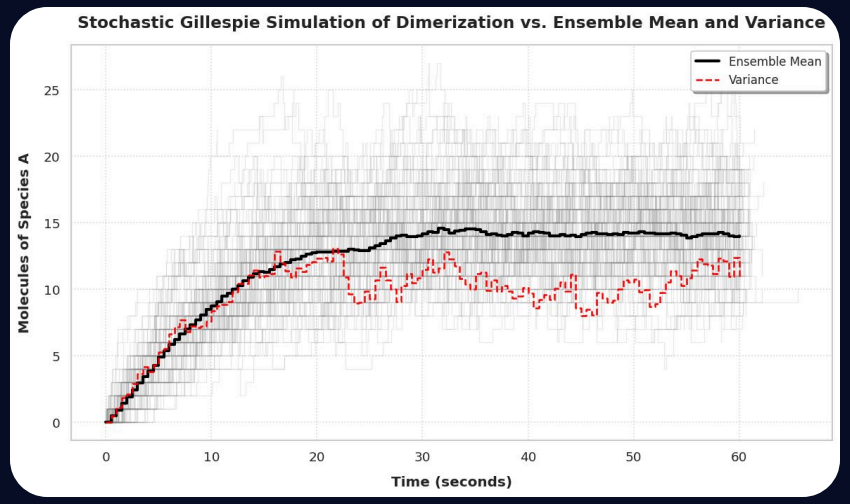
Since $\mathrm{Var}(n) < \langle n \rangle$, the system is sub-Poissonian. The quadratic decay term acts as a nonlinear feedback brake: upward fluctuations trigger stronger loss, narrowing the distribution around the mean.

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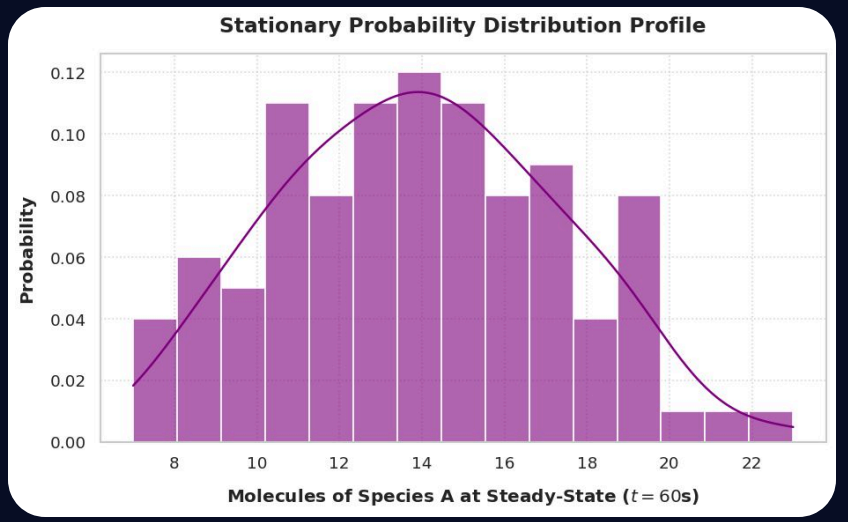
The Dimerization–Production Network: Sub-Poissonian Control

Nonlinear collision-driven decay changes the steady state from a Poisson distribution to a tightly regulated **sub-Poissonian** one.

STOCHASTIC RUN



STATIONARY DISTRIBUTION



Why it matters

Unlike linear systems, the reaction propensity scales quadratically with copy number, so higher states decay faster and the distribution becomes more concentrated than a Poisson baseline.

Conclusion: the network does not merely stabilize molecule counts — it enforces **sub-Poissonian variance** through nonlinear decay.

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Case Study: The Schlögl Model (Bistability & Trimolecular Kinetics)

The Schlögl model is a canonical example of nonlinear stochastic chemistry, showing how autocatalysis, decay, and constant influx can produce bistability and bimodal steady-state behavior.

The Model Framework

The Non-Linear System: A third-order autocatalytic network coupled with linear degradation and constant background influx.

$$3A \xrightarrow{k_1} 2A$$

$$2A \xrightarrow{k_2} 3A$$

$$A \xrightarrow{k_3}$$

$$\emptyset \xrightarrow{k_4} A$$

State Behavior: Tracks the discrete molecule population of species A over a 150-minute timeline, initializing from zero: $A(0) = 0$.

Multi-Particle Propensity Formulation

The system dynamics are governed by four competing propensity that determine the direction and frequency of state jumps.

- **Trimolecular Loss**: $\alpha_1 = k_1 \cdot A(A-1)(A-2)$ implies $A \leftarrow A - 1$
- **Bimolecular Gain**: $\alpha_2 = k_2 \cdot A(A-1)$ implies $A \leftarrow A + 1$
- **Linear Decay**: $\alpha_3 = k_3 \cdot A$ implies $A \leftarrow A - 1$
- **Baseline Creation**: $\alpha_4 = k_4$ implies $A \leftarrow A + 1$

$$a_0 = k_1 A(A-1)(A-2) + k_2 A(A-1) + k_3 A + k_4$$

Significance: The total propensity a_0 captures the full stochastic competition between creation and loss processes.

The Stationary Distribution & Recursive Detailed Balance

At steady state, the master equation reduces to local detailed balance between adjacent states, allowing the stationary distribution to be built recursively.

Master Equation Interpretation: The continuous-time probability profile $P_n(t)$ satisfies a one-step jump process, this explains how we obtain a recursive relation from the property.

$$P(n) = P(0) \prod_{i=1}^{n} \frac{\alpha_2(i-1) + \alpha_4}{\alpha_1(i) + \alpha_3(i)} = P(0) \prod_{i=1}^{n} \frac{k_2(i-1)(i-2) + k_4}{k_1 i(i-1)(i-2) + k_3 i}$$

Bistability: Fixed Points, Unstable Thresholds, & Bimodal Jumps

The macroscopic limit reveals a cubic rate law with three roots, separating the system into two stable basins and one unstable threshold.

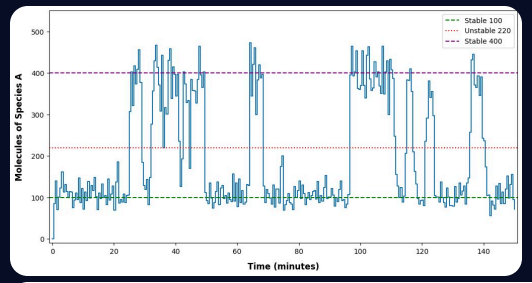
$$\frac{dA}{dt} = -k_1 A^3 + k_2 A^2 - k_3 A + k_4 = 0$$

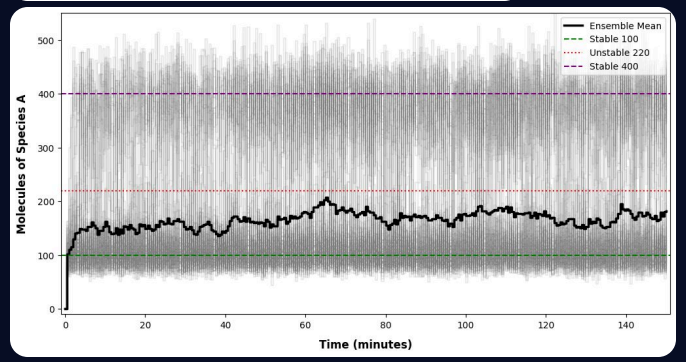
Three-Room Landscape

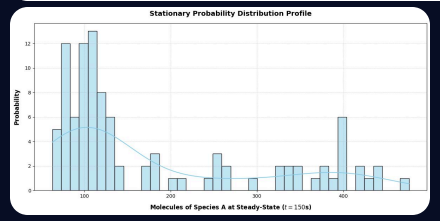
- $A_{s1} = 100$: Stable attractor, low-population basin.
- $A_u = 220$: Unstable saddle threshold; below this level trajectories relax to 100, above it they are driven toward 400.
- $A_{s2} = 400$: Stable attractor, high-population basin.

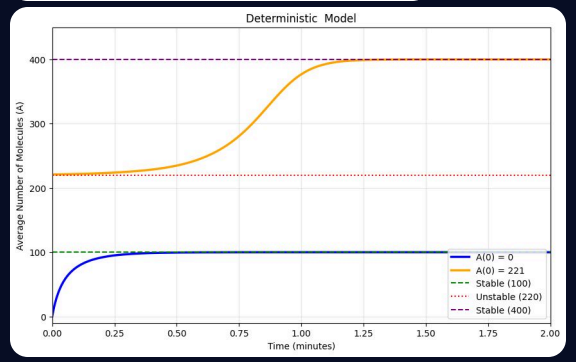
Stochastic Escapes & Bimodal Splitting: Starting from $A(0) = 0$, trajectories initially settle into the lower basin. Rare noise-driven bursts can cross the unstable threshold at 220, after which positive feedback rapidly transports the state toward 400, yielding a bimodal stationary distribution.

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Conclusion: Noise, Nonlinearity, and Bistability

The fundamental conclusion from the self-induced stochastic resonance model is that non-linear networks completely shatter classical deterministic assumptions. Because transitions occur in single-step (± 1) increments, the steady state satisfies localized detailed balance, defining a recursive propensity ratio that dictates a dramatic bimodal split into two stable probability peaks at 100 and 400 molecules separated by an unstable valley threshold at 220.

Bimodal Distribution

The stationary distribution separates into two dominant peaks at 100 and 400 molecules, with an unstable threshold at 220 acting as a separating valley between the two stable basins.

The Physical Reality

Although a macroscopic mean of approximately 180 can be computed, it is only a mathematical summary. The true behavior is stochastic: rare fluctuations trigger phase-like transitions that can move the system between low- and high-population states.

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The Schnakenberg Model: Bifurcations, Nullclines, and Noise

The Schnakenberg system captures how autocatalytic reaction dynamics can produce sharp phase-plane structure, bifurcation behavior, and noise-sensitive transitions.

1. Governing Deterministic System

The model describes an autocatalytic trimolecular reaction

$$2A + B \xrightarrow{k_1} 3A$$

$$\emptyset \underset{k_3}{\overset{k_2}{\rightleftharpoons}} A$$

$$\emptyset \xrightarrow{k_4} B$$

and tracks absolute molecular counts through the deterministic rate equations.

- $\frac{d\bar{A}}{dt} = K_1\bar{A}^2\bar{B} + K_2 - K_3\bar{A}$
- $\frac{d\bar{B}}{dt} = -K_1\bar{A}^2\bar{B} + K_4$

Here $K_1 = k_1/v^2$, $K_2 = k_2 v$, $K_3 = k_3$, and $K_4 = k_4 v$ are the volume-adjusted lumped constants.

2. Geometric Core: Nullclines and Peak Threshold

Setting the derivatives to zero reveals the phase-plane geometry. The nullclines define where each species stops changing and where the system's balance shifts.

- **$\bar{A}$-nullcline:** $\bar{B}(\bar{A}) = \frac{K_3\bar{A} - K_2}{K_1\bar{A}^2}$
- **$\bar{B}$-nullcline:** $\bar{B}(\bar{A}) = \frac{K_4}{K_1\bar{A}^2}$

The summit of the $\bar{A}$-nullcline occurs where $\frac{d\bar{B}}{d\bar{A}} = 0$, giving the critical boundary:

$$a_m = \frac{2K_2}{K_3} = 10 \quad \text{and} \quad b_m = \frac{K_3^2}{4K_1K_2} = 12,500$$

3. The Tipping Point Parameter

The threshold k_b (or γ) marks the exact point where the $\bar{B}$-nullcline intersects the summit (a_m, b_m) of the $\bar{A}$-nullcline.

$$k_b \approx a_m^2 \cdot b_m \cdot K_1 = 50\ \text{s}^{-1}$$

This value separates stable quiescence from self-sustained oscillatory behavior, making it the key bifurcation boundary in the model.

4. Key Takeaways: Deterministic vs. Stochastic Dynamics

1. **Noise-induced bifurcation blurring:** Deterministically, the switch at k_b is sharp. Stochastically, intrinsic fluctuations can trigger periodic motion even below the threshold.
2. **Phase desynchronization:** Individual Gillespie paths remain oscillatory, but random timing shifts make ensemble averages look damped or fading.

Deterministic and stochastic regimes

Parameter Regime	Phase Plane Interaction	Deterministic Output	Stochastic Output (Gillespie)
$K_4 < k_b$	Intersection on the stable left branch.	Stable steady state; the system settles to a dead stop.	**Excitability:** Quiet baseline, but noise can push paths past a_m, sparking isolated action potentials.
$K_4 > k_b$	Intersection on the unstable right branch.	**Bifurcation:** The steady state breaks into a permanent **Oscilation**	Continuous, sustained rhythmic oscillations.

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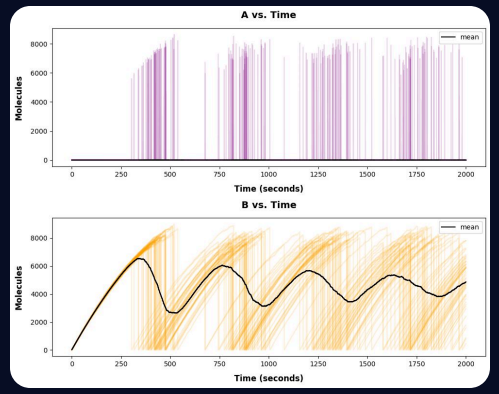
Bifurcation Analysis: Parameter Sensitivity in the Schnakenberg Model

This page compares the Schnakenberg model under two parameter regimes, **k4 = 25** and **k4 = 100**, to show how changing a single rate parameter alters the phase-plane behavior and the transition between steady-state and oscillatory dynamics.

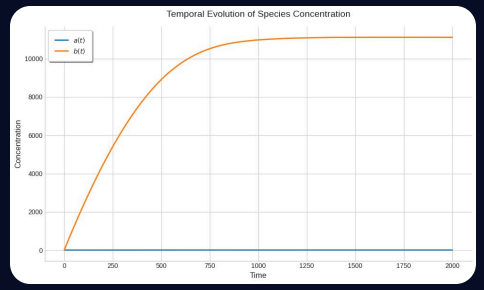
DETERMINISTIC VS. STOCHASTIC COMPARISON | PARAMETER SENSITIVITY

Low forcing regime: $k_4 = 25$

What the images show: The left column compares the model behavior when k_4 is below the bifurcation threshold. The deterministic and stochastic views highlight how the system remains near a stable state, while noise can still trigger transient excursions.



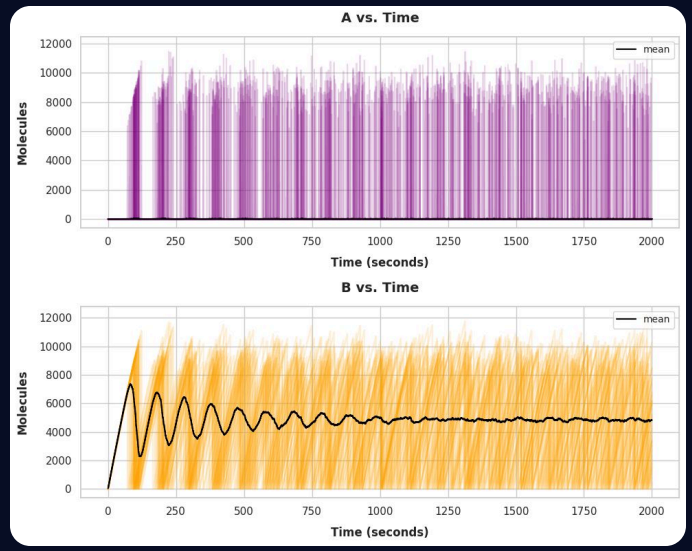
Caption: Phase-plane behavior and trajectory response for $k_4 = 25$, illustrating the lower-parameter regime where the system remains closer to the stable branch.



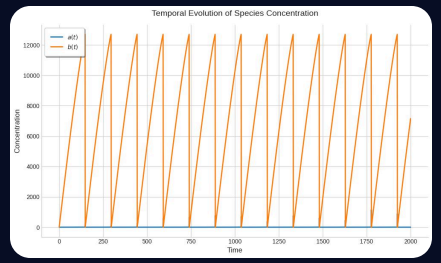
Caption: Corresponding deterministic-based visualization for $k_4 = 25$, showing how fluctuations can perturb the deterministic behavior.

High forcing regime: $k_4 = 100$

What the images show: The right column compares the model behavior when k_4 exceeds the bifurcation threshold. In this regime, the phase-plane geometry shifts and sustained oscillatory behavior becomes more apparent.



Caption: Phase-plane behavior and trajectory response for $k_4 = 100$, showing the system on the post-bifurcation side of the threshold.



Caption: Corresponding deterministic trajectory-based visualization for $k_4 = 100$, emphasizing the sustained rhythmic dynamics in the higher-parameter regime.

Key Summary

Deterministically, the Schnakenberg model predicts a clean, binary switch between steady states and limit cycles at a sharp bifurcation threshold. With stochastic molecular noise, however, this boundary becomes blurred: random fluctuations can induce premature rhythmic oscillations below the expected threshold and cause independent trajectories to lose phase synchronization over time, smoothing the transition between qualitatively distinct dynamic states.

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Case Study: Stochastic Focusing in Biochemical Systems

How ultra-low-copy intermediates can cause stochastic fluctuations to amplify downstream responses far beyond deterministic predictions.

1. System Overview & Reaction Network

An indirect coupling network where signal molecule A regulates product B through the ultra-low-copy intermediate species C.

$$\emptyset \xrightarrow{k_1} C \xrightarrow{k_2} B \xrightarrow{k_3} \emptyset$$

$$A + C \xrightarrow{k_4} A$$

$$\emptyset \xrightarrow{k_5} A \xrightarrow{k_6} \emptyset$$

The intermediate C acts as the critical bottleneck, making the downstream response highly sensitive to molecular noise.

2. Mathematical Modeling in Molecular Numbers

Let $\bar{A}$, $\bar{B}$, and $\bar{C}$ denote the approximate molecule counts in a reactor of volume ν.

$$\frac{d\bar{A}}{dt} = k_5\nu - k_6\bar{A}$$

$$\frac{d\bar{B}}{dt} = k_2\bar{C} - k_3\bar{B}$$

$$\frac{d\bar{C}}{dt} = k_1\nu - k_2\bar{C} - \frac{k_4}{\nu}\bar{A}\bar{C}$$

These mean-field equations provide the deterministic baseline used to contrast against stochastic simulation outcomes.

3. Signal Variance & Poisson Probability Distribution

Because signal A evolves through an independent birth-death process, its steady-state profile follows a classical **Poisson distribution**. These properties can be derived from the Chemical Master Equation.

$$P(N_A) = \frac{\langle A \rangle^{N_A} e^{-\langle A \rangle}}{N_A!}$$

- **Variance property:** A fundamental constraint of this distribution is that its variance exactly tracks its mean ($\mathrm{Var}(A) = \langle A \rangle$).

4. The Core Phenomenon: Stochastic Focusing

When signal A drops from $10 \to 5$ molecules, the response of product B diverges sharply between deterministic and stochastic descriptions.

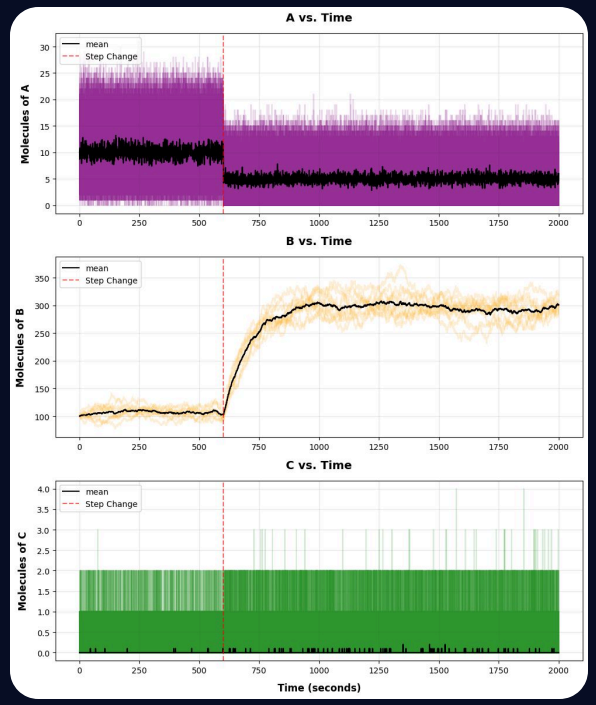
- **Deterministic view:** Predicts a steady-state product increase from $100 \to 198$ molecules (or exactly 200 under ideal limits).
- **Stochastic view:** The SSA mean of B rises from $100 \to\sim 300$ molecules.
- **Interpretation:** Because C exists in only a few discrete states (0, 1, 2 molecules), rapid fluctuations nonlinearly amplify production, causing classical ODEs to underpredict output by over **33%**.

This is a textbook case of stochastic focusing: noise does not merely perturb the response, it reshapes the effective input-output relationship of the system.

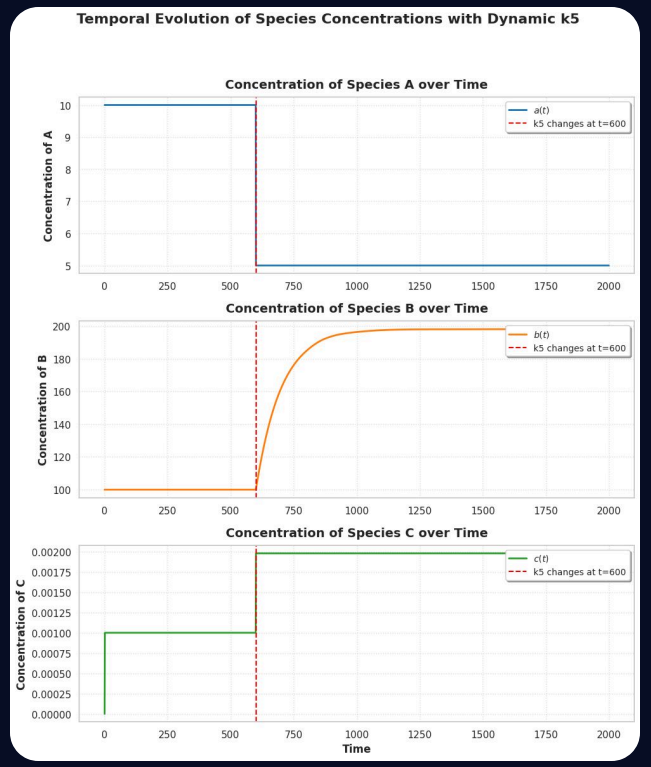
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Stochastic focusing shows how randomness in ultra-low-copy intermediates can amplify downstream responses instead of simply adding variation.

STOCHASTIC RUN



DETERMINISTIC RUN



Ultra-low-copy intermediates

These molecules often fluctuate between zero, one, or two copies, making the system highly sensitive to random events.

Fluctuations amplify response

Noise can let the intermediate escape degradation and spike when the upstream signal falls, boosting downstream output.

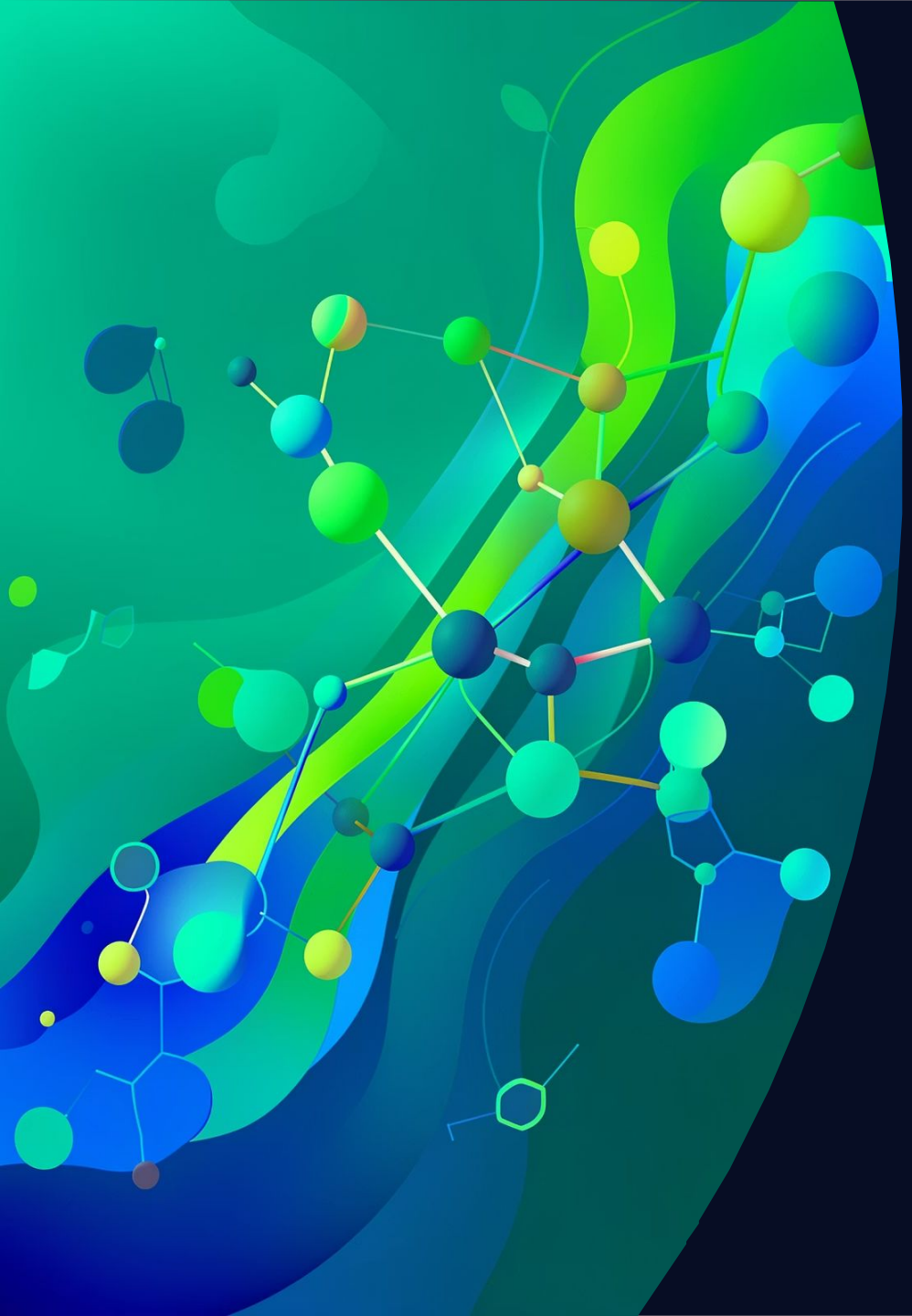
Deterministic models miss it

By smoothing over discrete molecular events, they underestimate the strength of the downstream response.

Key insight

In small molecular systems, randomness is not just error—it can become a driver of function.

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The Power of Stochastic Simulation

- Noise can shape molecular behavior, especially in small-copy systems where random fluctuations matter most.
- The Gillespie Algorithm provides an exact way to simulate reaction events without smoothing away discreteness.
- Nonlinearity can amplify randomness, producing behaviors that deterministic models may miss.
- Understanding stochastic processes is essential for interpreting and predicting biochemical dynamics realistically.

Closing thought: In biochemistry, stochasticity is not just a complication—it is often a fundamental part of how systems function.

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